St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | DLA

Vertebral Column & Spinal Cord (DLA)

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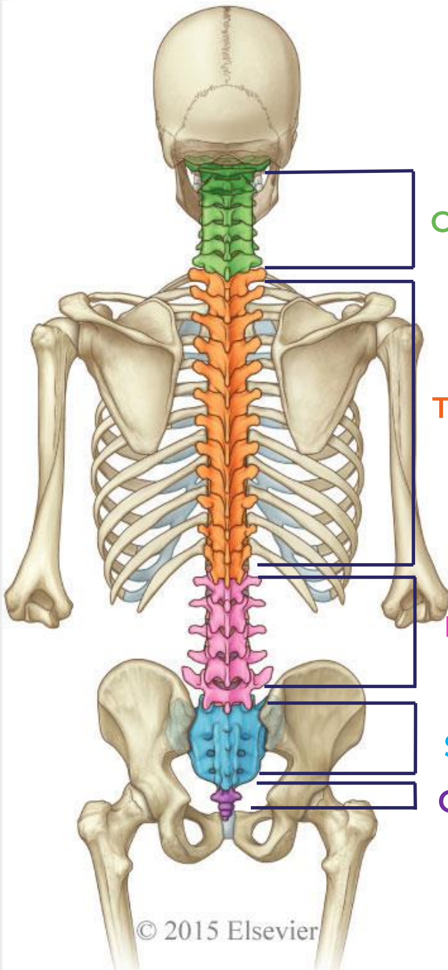
Assigned Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 4th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 2: pg 67- 86
- Online Textbook: Chapter 2 https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323393041000026

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1043	Describe the different components of a typical vertebra, highlighting the features that are unique to each region.
SOM.MKII.BPM1.1.MSK.1.ANAT.1044	Describe the joints found between and associated movements of the typical vertebrae.
SOM.MKII.BPM1.1.MSK.1.ANAT.1045	Describe the structure of the atlas and the axis.
SOM.MKII.BPM1.1.MSK.1.ANAT.1046	Describe the structure of the intervertebral disc.
SOM.MKII.BPM1.1.MSK.1.ANAT.1047	Describe the joints between the atlas, axis and skull, including the movements possible at each.
SOM.MKII.BPM1.1.MSK.1.ANAT.1048	Describe the function of the transverse and alar ligaments of the atlas.
SOM.MKII.BPM1.1.MSK.1.ANAT.1049	Describe the five ligaments that support the vertebral column.
SOM.MKII.BPM1.1.MSK.1.ANAT.1050	Describe the normal curvatures of the vertebral column and their development.
SOM.MKII.BPM1.1.MSK.1.ANAT.1051	Describe scoliosis, kyphosis and lordosis
SOM.MKII.BPM1.1.MSK.1.ANAT.1052	Describe the difference between spinal gray matter and spinal white matter with respect to location, composition and appearance (Review from FTM)
SOM.MKII.BPM1.1.MSK.1.ANAT.1053	Describe the external shape of the spinal cord and the reason for its different appearance in some locations (Review from FTM)
SOM.MKII.BPM1.1.MSK.1.ANAT.1054	Describe the organization and appearance of gray matter in the different regions of the spinal cord (Review from FTM)
SOM.MKII.BPM1.1.MSK.1.ANAT.1055	Describe the functional modalities associated with the ventral, dorsal and lateral gray horns (Review from FTM)
SOM.MKII.BPM1.1.MSK.1.ANAT.1056	Describe the extent, attachments and spaces of the spinal meninges.
SOM.MKII.BPM1.1.MSK.1.ANAT.1057	Describe the extent and the contents of the dural sac (lumbar cistern).
SOM.MKII.BPM1.1.MSK.1.ANAT.1058	Describe the extent and the attachments of filum terminale and denticulate ligaments.
SOM.MKII.BPM1.1.MSK.1.ANAT.1059	Describe the changes in vertebral level of the conus medullaris from 8 weeks to 24 weeks to newborn to adult.
SOM.MKII.BPM1.1.MSK.1.ANAT.1060	Describe at which vertebral level a lumbar puncture (spinal tap) should be performed in an infant/child versus in an adult, and why.
SOM.MKII.BPM1.1.MSK.1.ANAT.1061	List the structures the needle would pierce in a lumbar puncture.
SOM.MKII.BPM1.1.MSK.1.ANAT.1062	Describe the location and function of the internal vertebral venous plexus (of Batson) and its role in the spread of infections and/or cancer.
SOM.MKII.BPM1.1.MSK.1.ANAT.1063	Describe the blood supply and venous drainage of the spinal cord and its clinical significance.

Vertebral Column



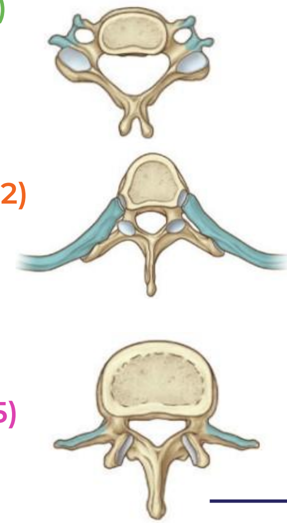
Cervical Vertebrae (7)

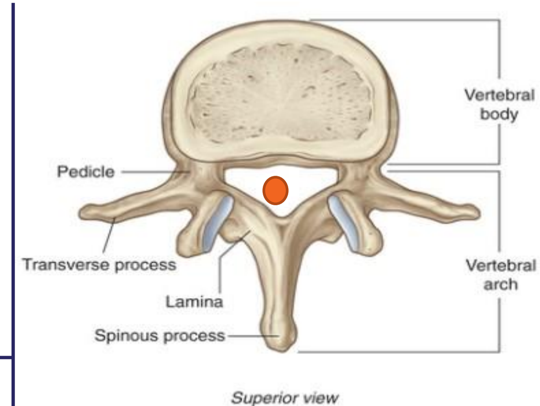
Thoracic Vertebrae (12)

Lumbar Vertebrae (5)

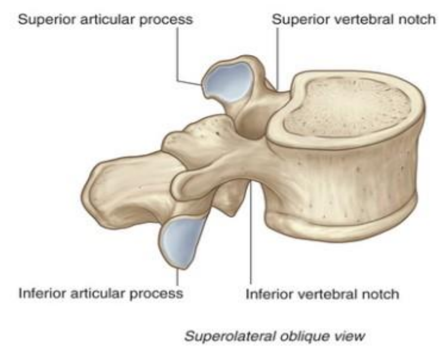
Sacrum (5)

Coccyx (3 -5)

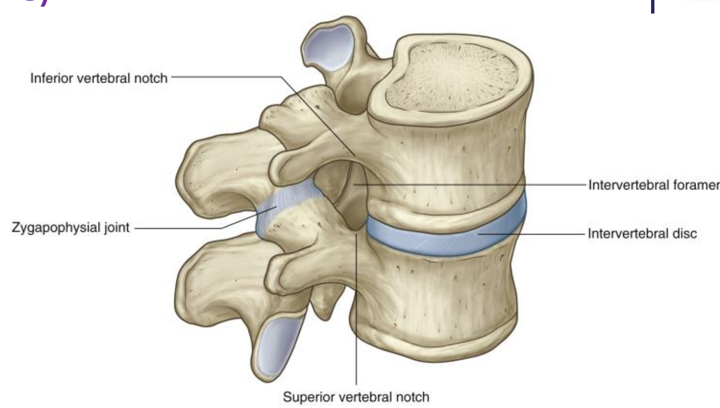




Superior view

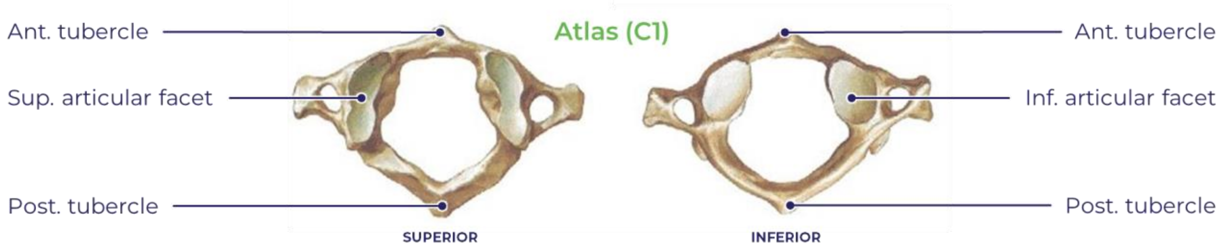


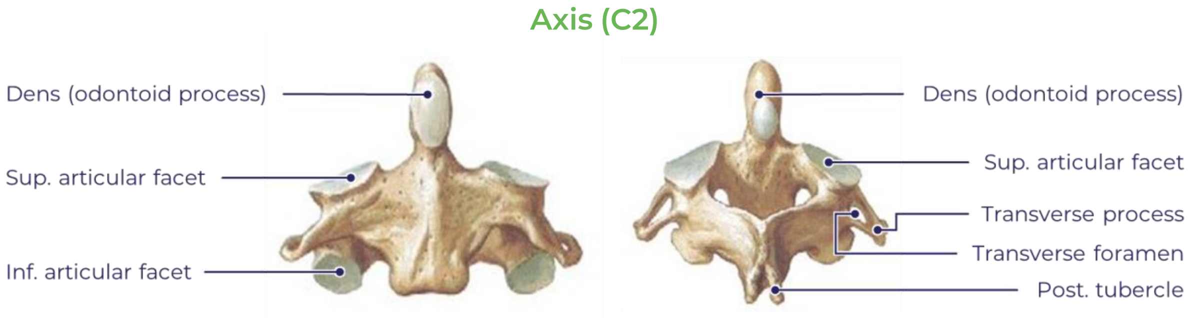
Superolateral oblique view

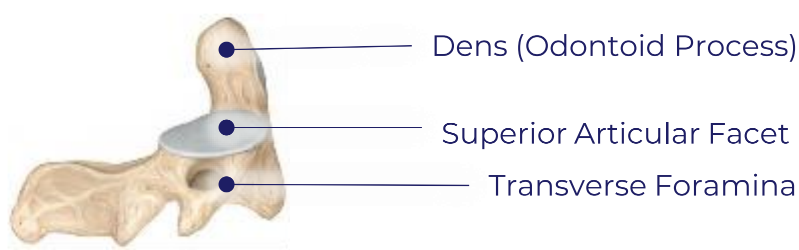


Cervical Vertebrae

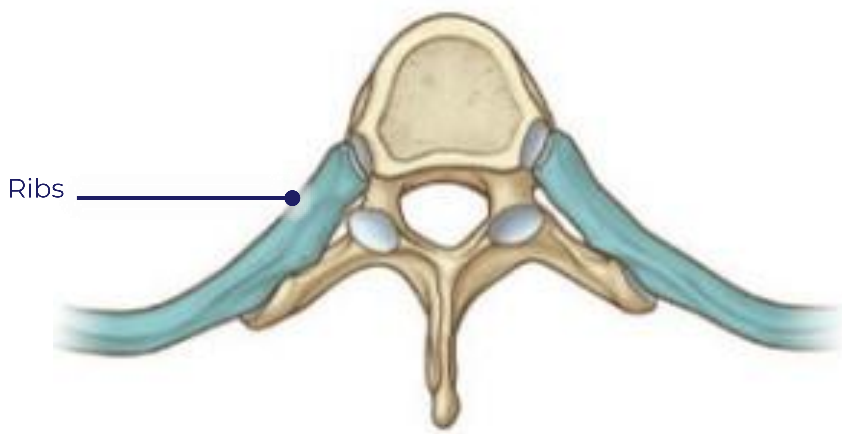
- Transverse foramina
- Bifid spinous process except C1 and C7
- Uncinate process
- Small vertebral body
- Large vertebral foramen

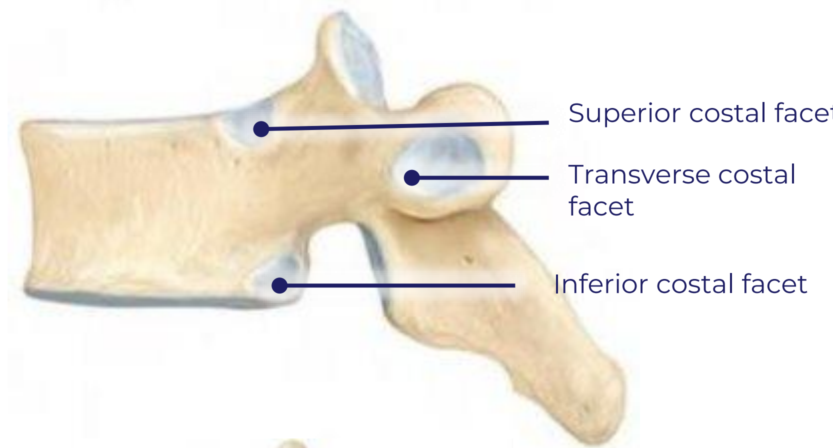


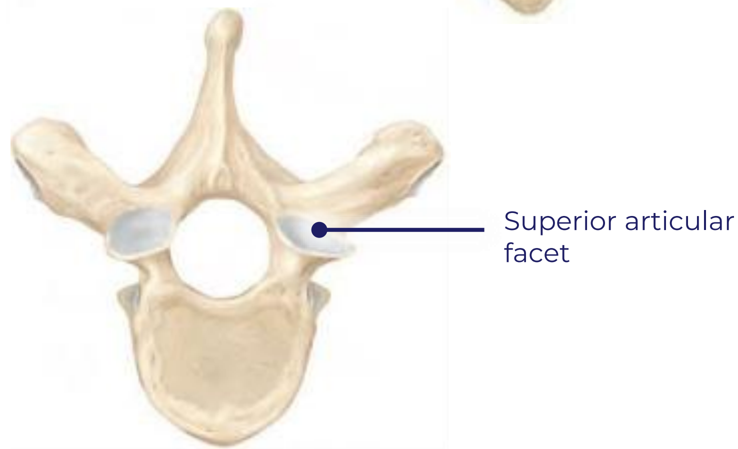




Thoracic Vertebrae

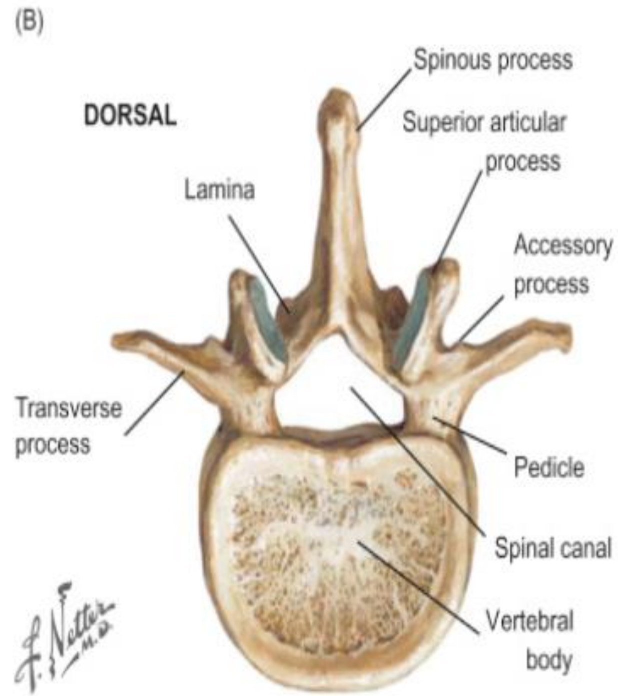






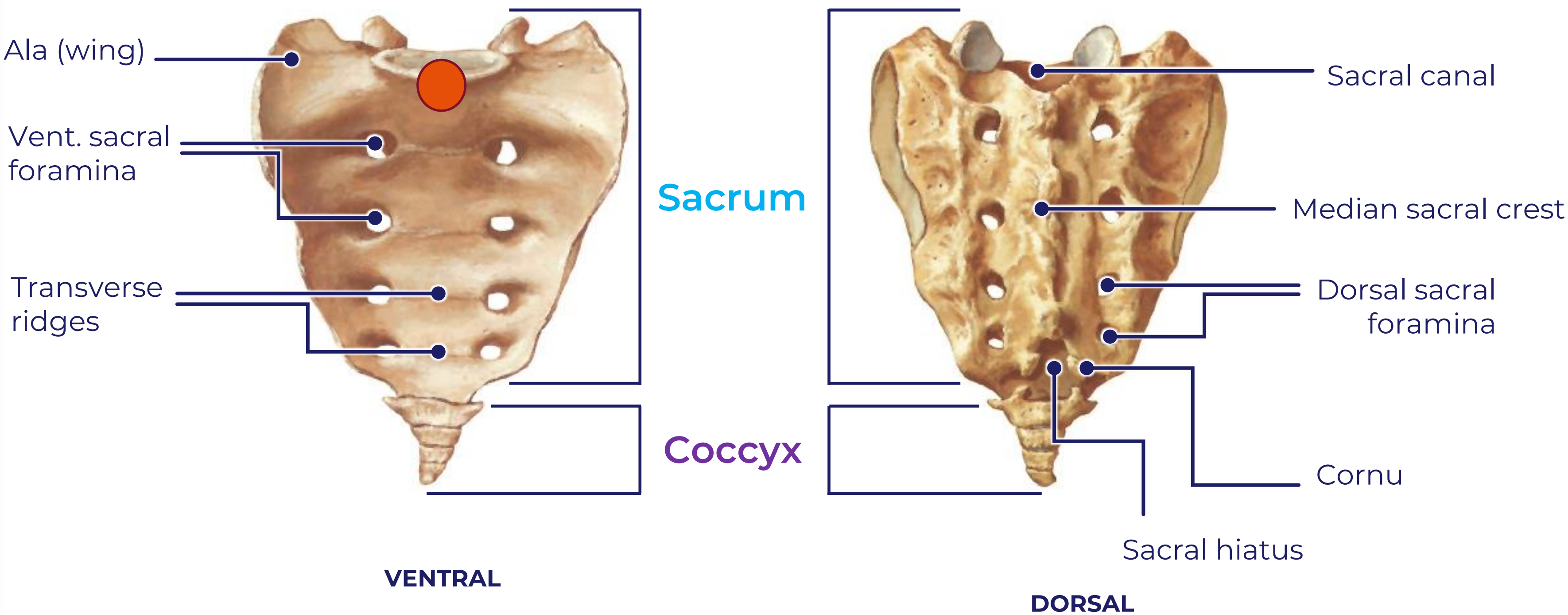
- Superior and inferior costal facets along posterior-lateral vertebral body
- Transverse costal facets
- Long spinous process oriented inferiorly
- Heart shaped vertebral body

Lumbar Vertebrae



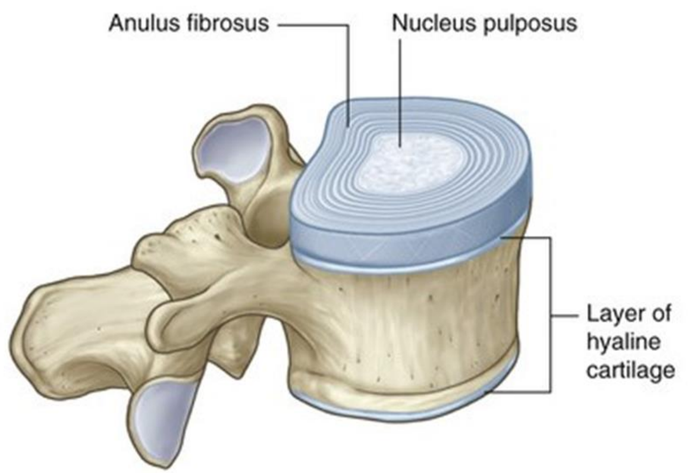
- Largest and thick vertebral body
- Short horizontal transverse process
- Short spinous process
- Triangular vertebral foramen
- Mamillary processes originates from the posterolateral margin of the superior articular process
- Accessory processes tubercle on the dorsal aspect of the base of the lumbar transverse process

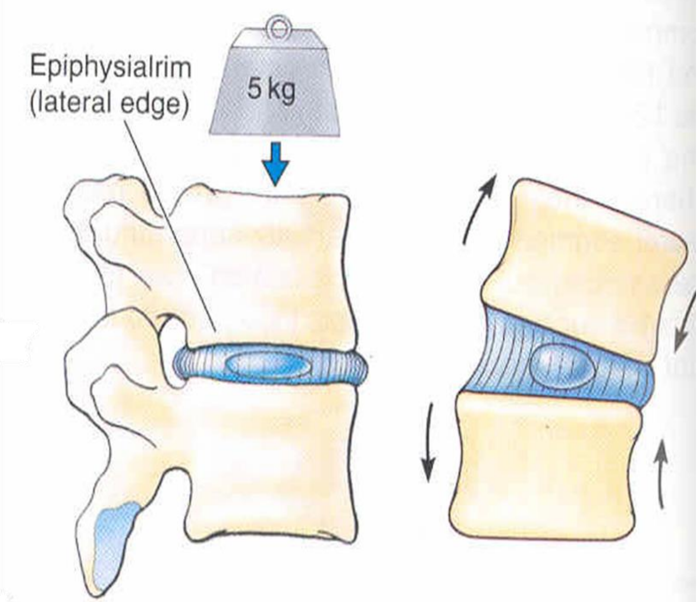
Sacrum and Coccyx



- Fused bones
- L-shaped facet on lateral ala that articulates with hip bone (Sacroiliac Joint)
- Spinal nerves exit from sacral foramina
- **Sacral Promontory** best visualized from the lateral view. Radiological landmark!

Intervertebral Disc





(C) Lateral view of disc when erect (weight bearing)

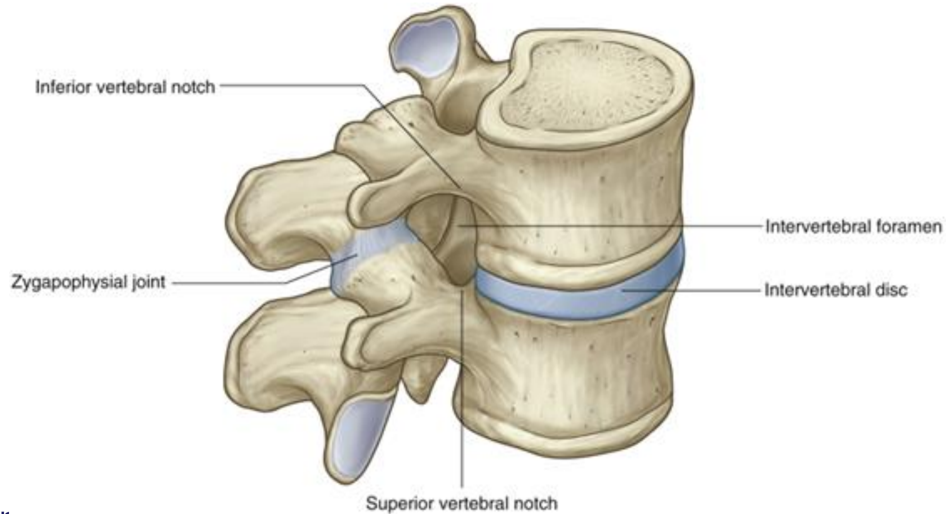
(D) Anterior view during lateral flexion

- Assist in stabilizing the vertebral column
- Weight bearing support
- Adds flexibility to the spine
- Facilitates the curvature of the spine
- Outer anulus fibrous surrounding an inner/central nucleus pulposus (**a remnant of the notochord**)

Joints of the Spine

- **Stabilizers**
  - **Fibrocartilaginous Joints**

- **Movers**
  - **Facet Joints (zygapophyseal joint)**; formed between superior and inferior articular processes
  - **Costovertebral Joints**; formed between thoracic vertebral body costal facet and costal facet of the ribs head
  - **Costotransverse Joints**; formed between thoracic rib and the transverse process costal facet of the thoracic vertebrae



Fibrocartilaginous Joints and Facet Joints

Costovertebral and Costotransverse Joints

Joints of the Spine

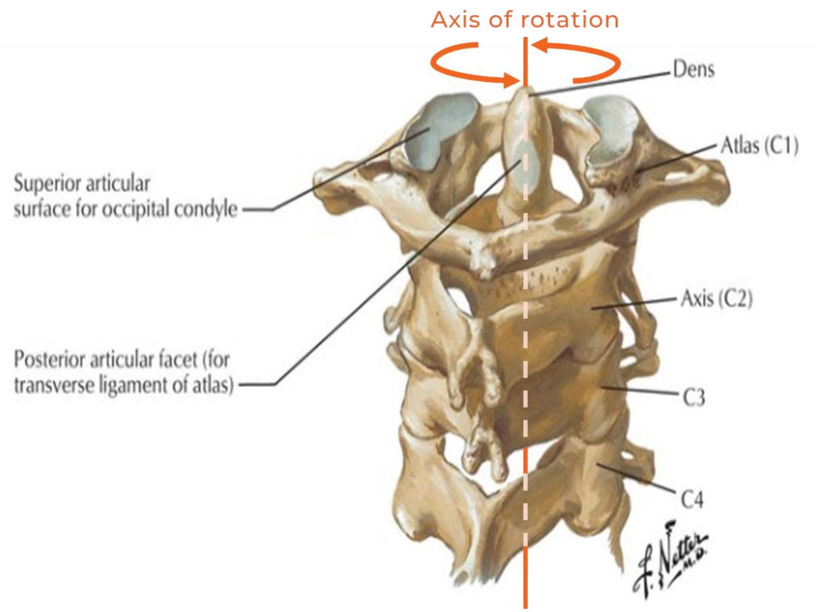
- Movers Continued
  - **Atlantooccipital Joint**; formed between the C1 vertebra, superior articular facet and condyles of the spine. Responsible for the "**yes" motion of head**.
  - **Atlantoaxial Joint**; 3-point articulation, formed between the C1 vertebrae, inferior articular process and superior articular process of C2 AND C1 vertebrae anterior arch with the Odontoid process of C2. Responsible for the "**No" motion of head**.

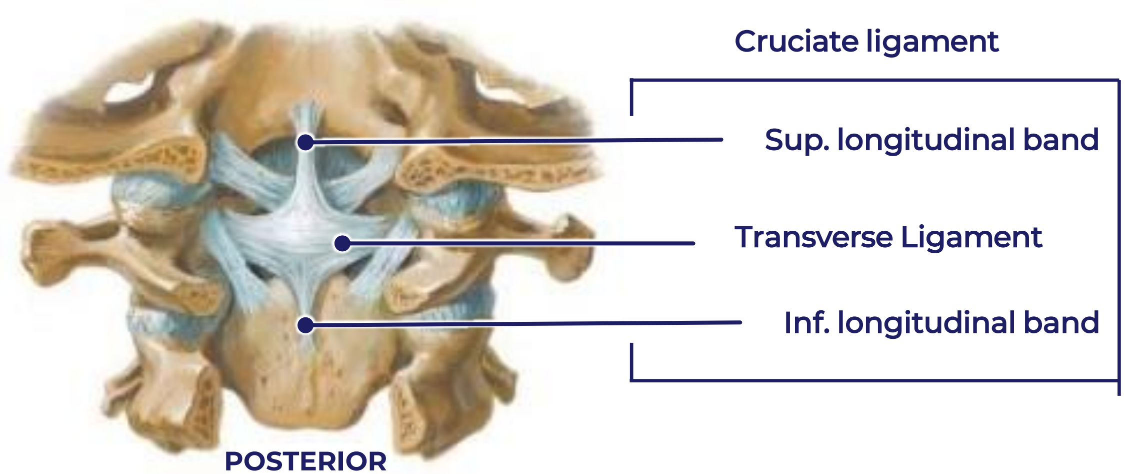
Atlantooccipital Joint

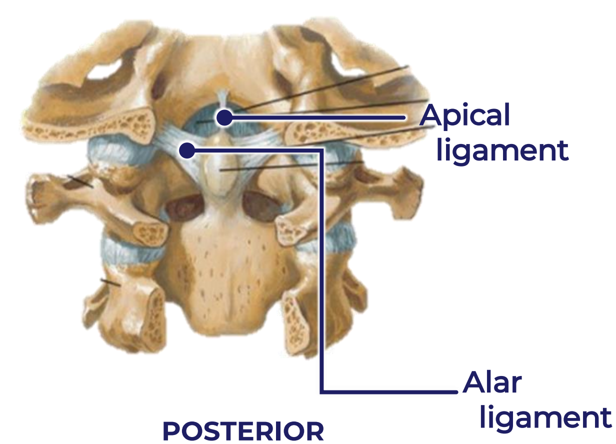
Atlantoaxial Joint

Ligamentous Support (Cervical)

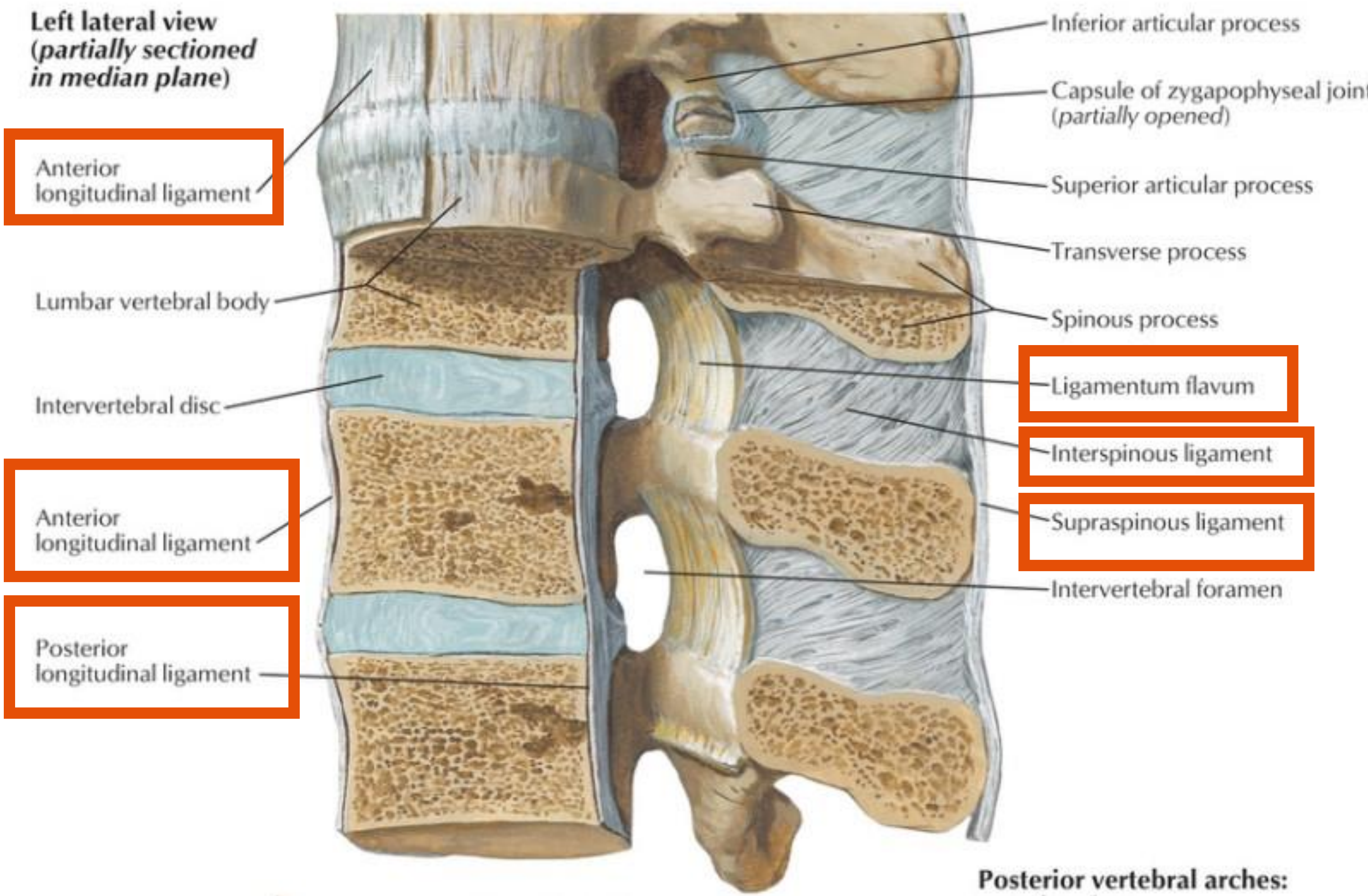
- **Cruciate Ligament** helps to hold dens in place during rotation of head. Superior and inferior longitudinal bands pass from the transverse ligament to occipital bone superiorly, and inferiorly to body of C2
- **Transverse Ligament** horizontal component of cruciate ligament. Strongest and thickest ligament that supports atlantoaxial joint
- **Apical Ligament** located deep to cruciate ligament, attaching dens to skull
- **Alar Ligament** runs laterally from dens to skull





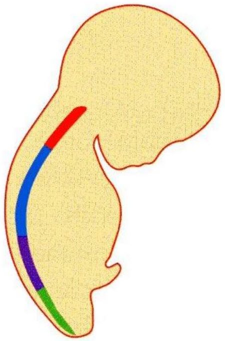


Ligamentous Support



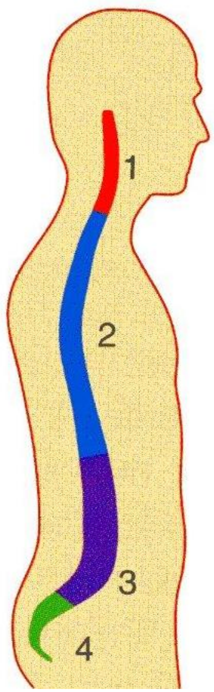
- Anterior longitudinal ligament prevents hyper- extension
- Posterior longitudinal ligament, interspinous ligament and supraspinous ligament prevents hyperflexion
  - Nuchal ligament (ligamentum nuchae) located in cervical region also prevents hyperflexion
- Ligamentum Flavum assist with repositioning of the vertebral column from the flexed position

Spine Curvatures



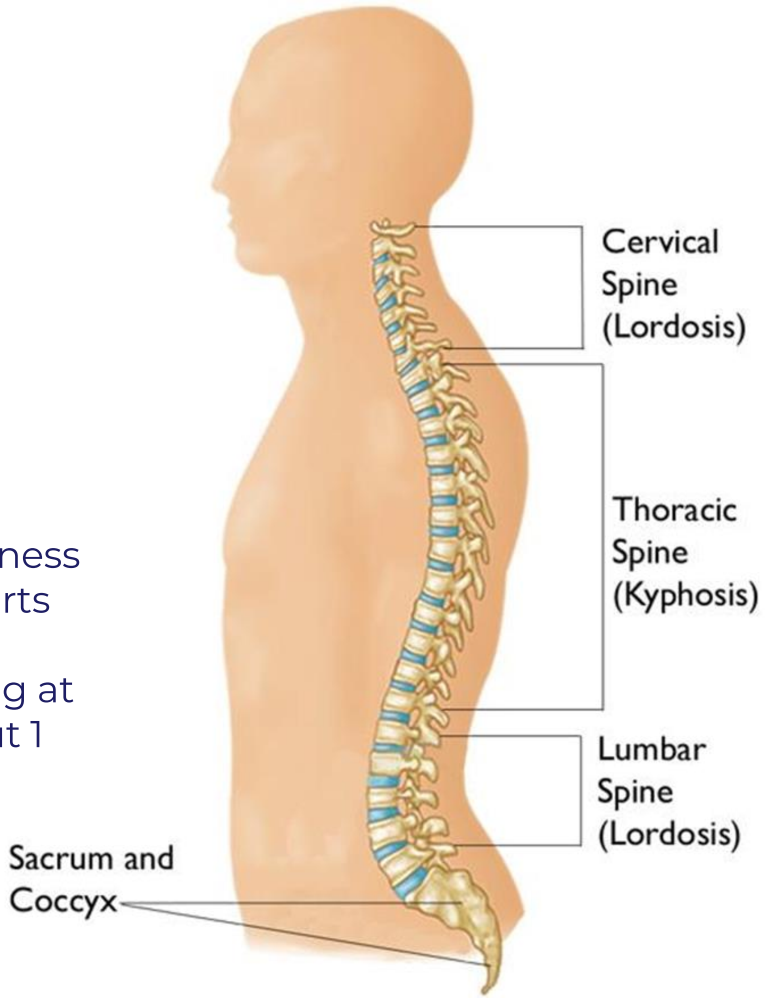
Primary curvatures

- Develop during fetal period in relation to the fetal (flexed) position

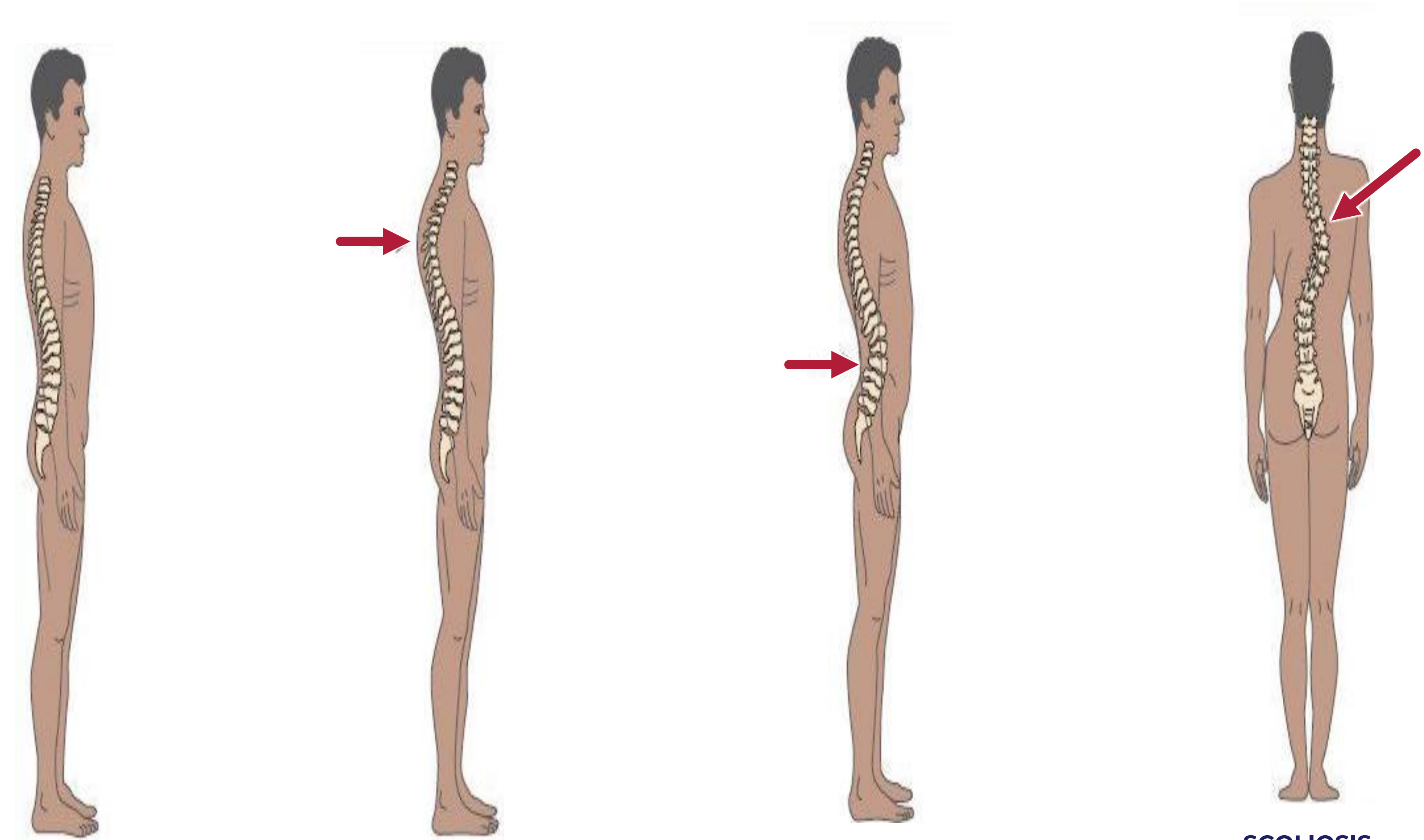


Secondary curvatures

- Result from extension from flexed fetal position
- Maintained by differences in thickness between anterior and posterior parts of IV discs
- Control of head (1-2 months), sitting at about 6 months and walking about 1 year old



Spine Curvature Disorders



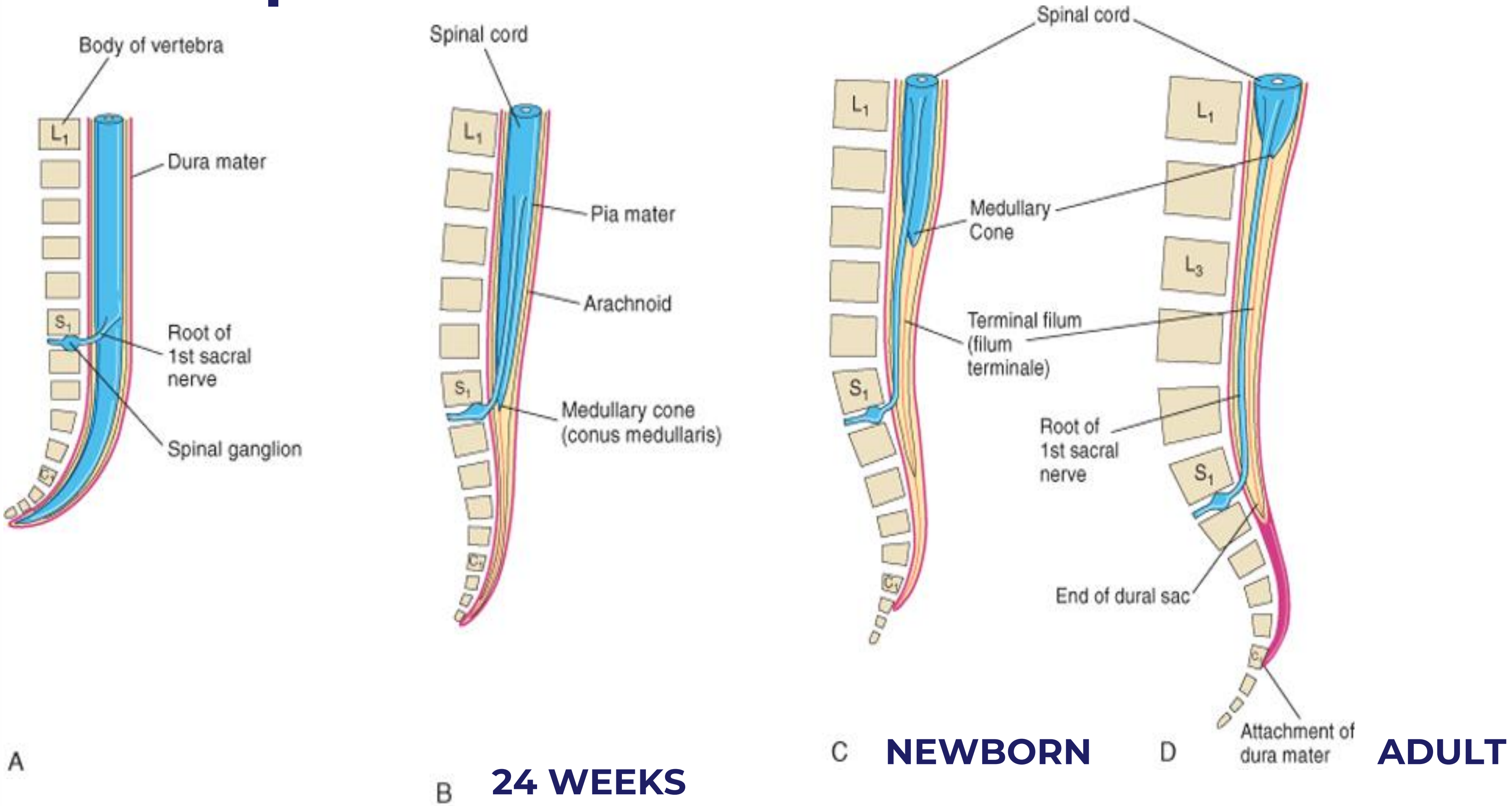
NORMAL

KYPHOSIS
HYPERKYPHOSIS

LORDOSIS
HYPERLORDOSIS

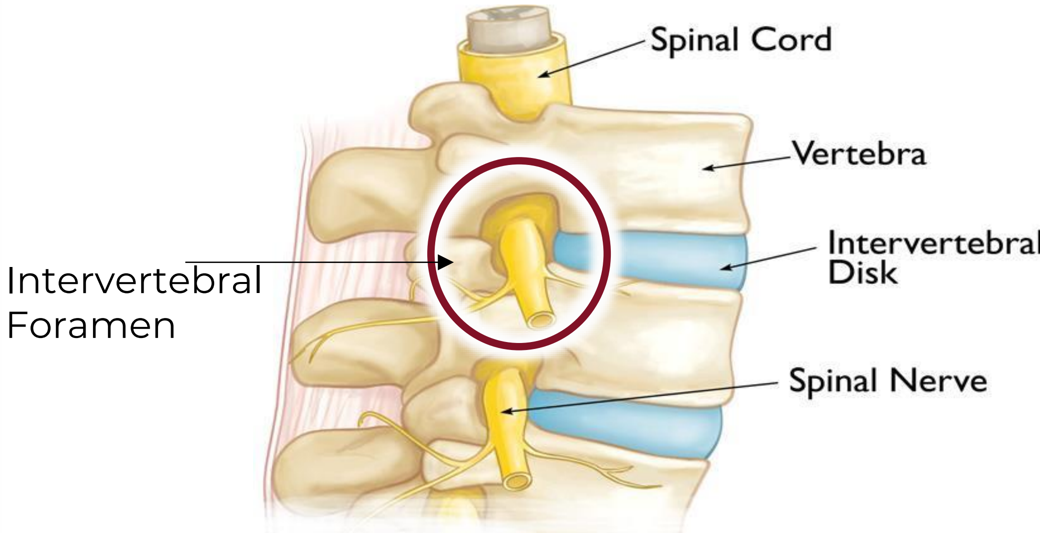
SCOLIOSIS

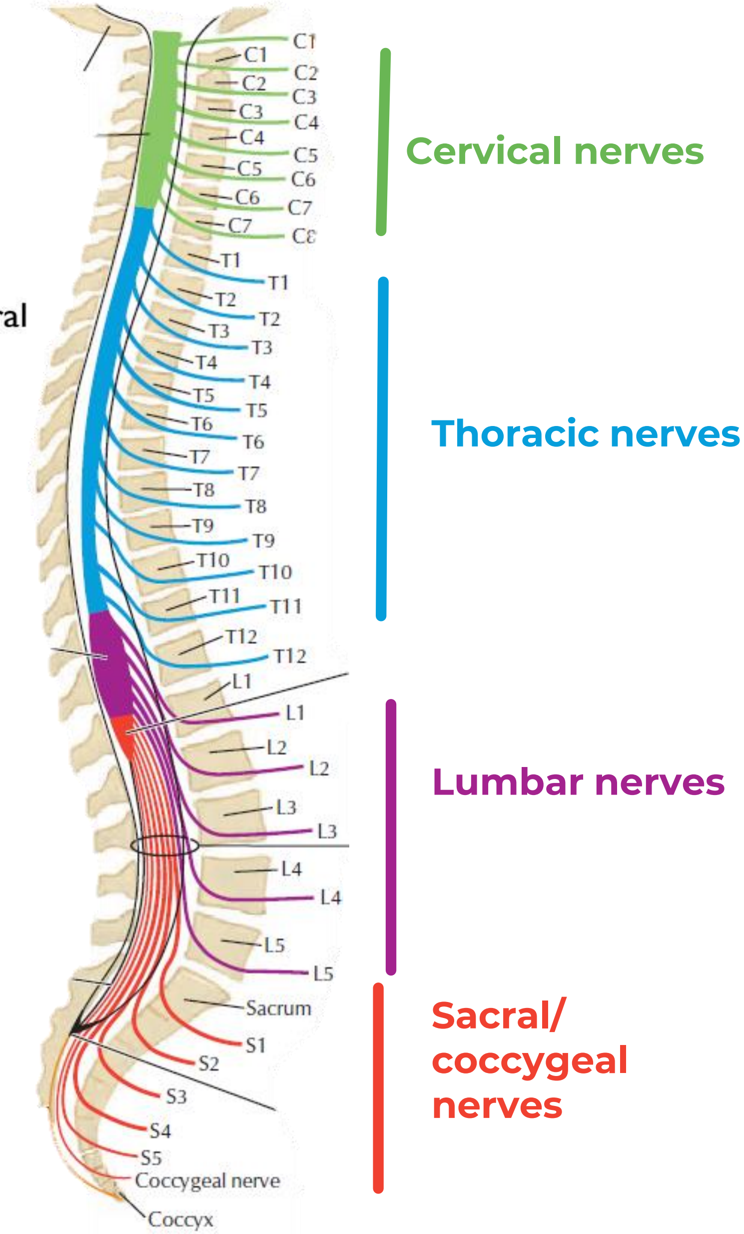
Spinal Cord and Vertebral Column Development



- Differential in growth rate of meninges and vertebrae results in the spinal cord not spanning the full length of the vertebral canal
- Newborn and adult have long vertebral column and short spinal cord
  - Newborn conus medullaris ends at L3/L4
  - Adult Conus medullaris ends at L1/L2

Spinal Cord Segments and Spinal Nerves

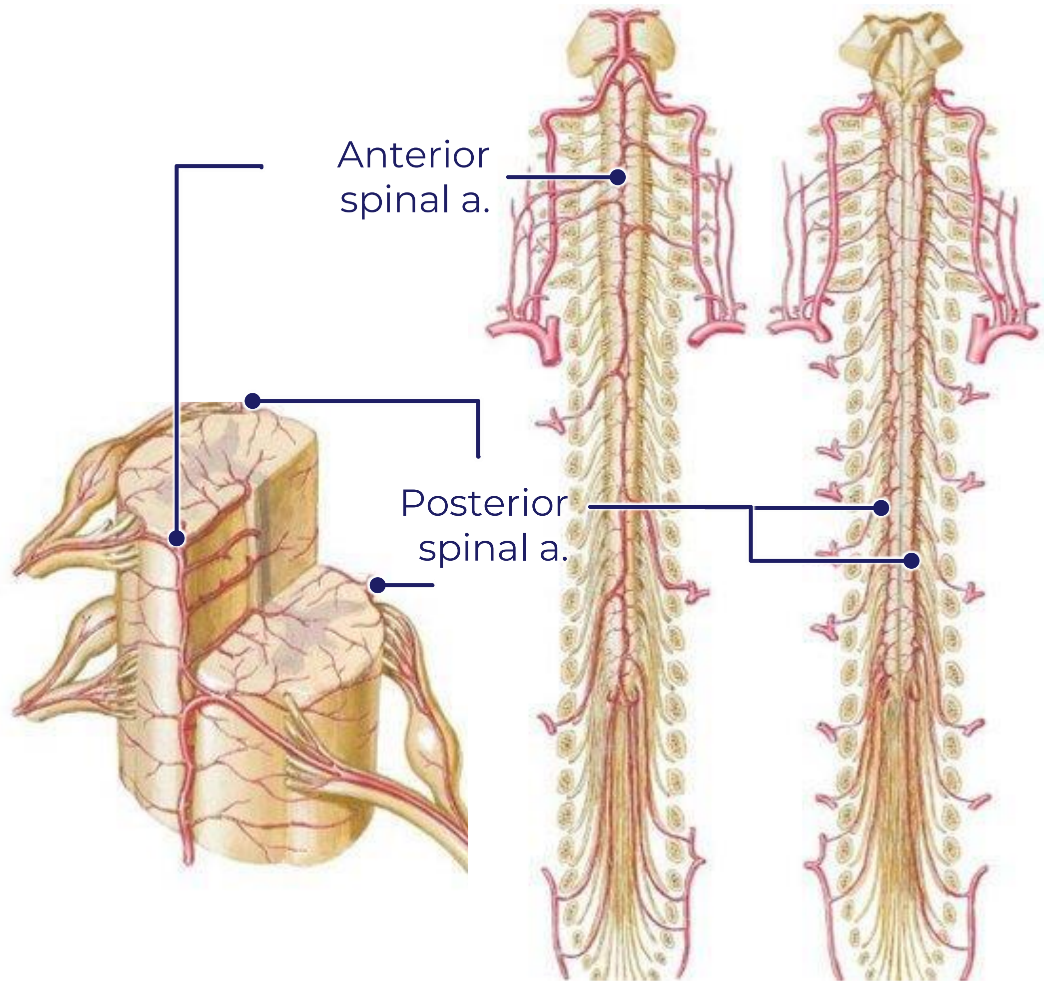




- Cervical, Thoracic, and Lumbar spinal nerves exit through interverbal foramina
- C1 – C7 Spinal nerves exit above their respective vertebrae. C8 spinal nerve exits below the C7 vertebra
- Thoracic and Lumbar spinal nerves exit below their named vertebrae

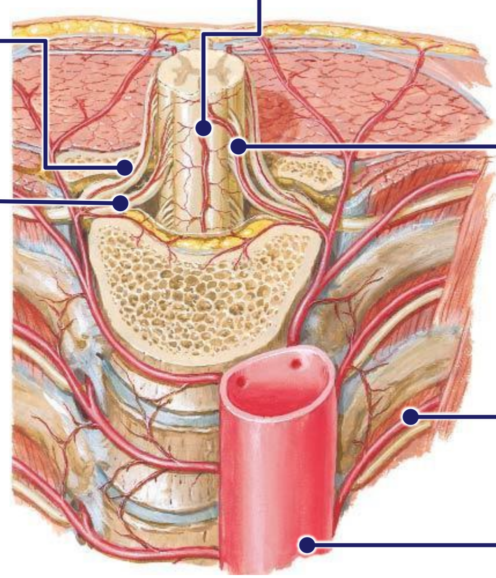
Blood Supply Spinal Cord

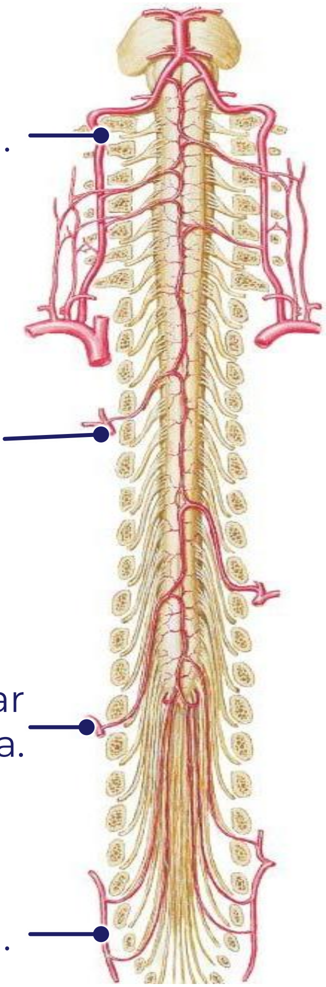
- Location: one anterior and two posterior spinal arteries.
- The Anterior Longitudinal or Anterior Spinal Artery originates from vertebral artery
  - **Supplies Anterior 2/3 of spinal cord**
  - **Anterior and Lateral horns and columns**
- The Posterior Longitudinal Artery or Posterior Spinal Artery originates mostly from the Posterior Inferior Cerebellar Artery.
  - **Supplies posterior 1/3 of spinal cord**
  - **Posterior Horns and columns**



Blood Supply Spinal Cord

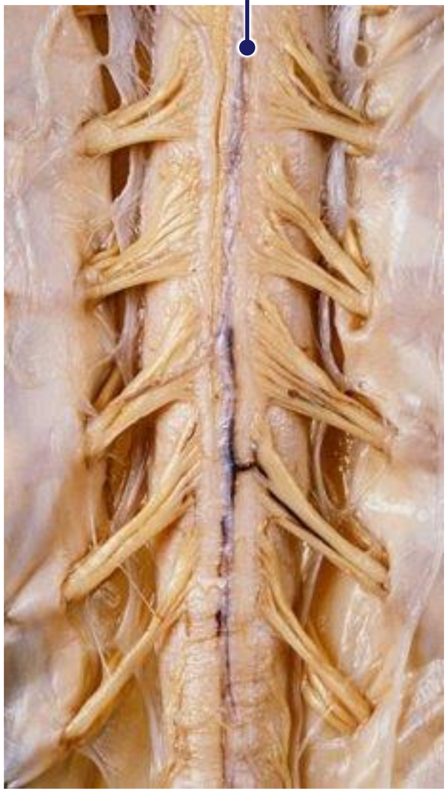
- Feeder arteries entering via IV foramina
- Originate from body wall arteries at each segment of spinal cord
- Two types of segmental blood supply:
  - **Segmental medullary arteries** which anastomose with longitudinal spinal arteries
  - **Segmental Radicular branches** that supply dorsal and ventral roots
- Anterior and posterior spinal veins drain spinal cord. Veins drain into **Batson's Plexus located in epidural space**

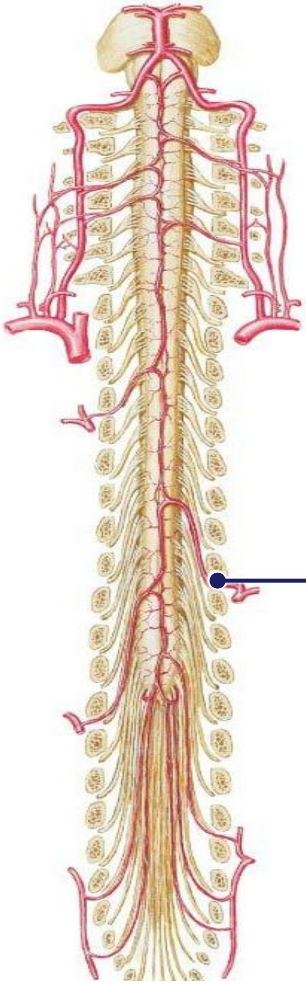




Blood Supply Spinal Cord: Artery of Adamkiewicz

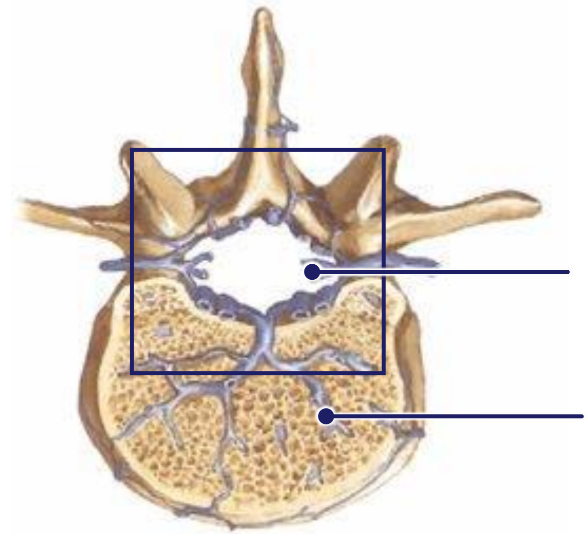
- Great anterior segmental medullary artery(Artery of Adamkiewicz)
- Largest segmental medullary artery
- Arises from lower thoracic or upper lumbar region
- Important source of blood supply to lower lumbar and sacral parts (conus medullaris) of spinal cord
- Spinal cord circulatory impairment ensues if there is damage to segmental medullary arteries

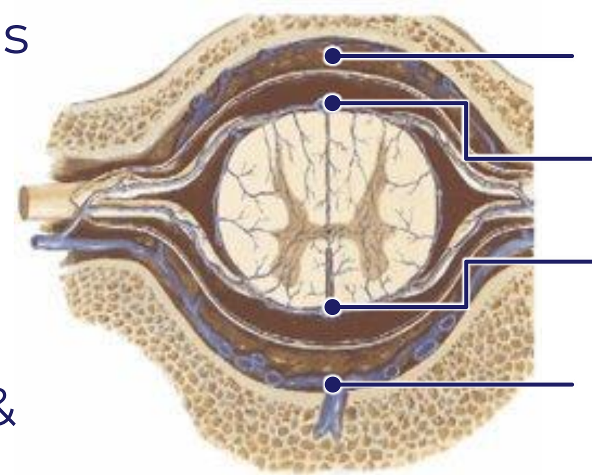




Venous Drainage of Spinal Cord

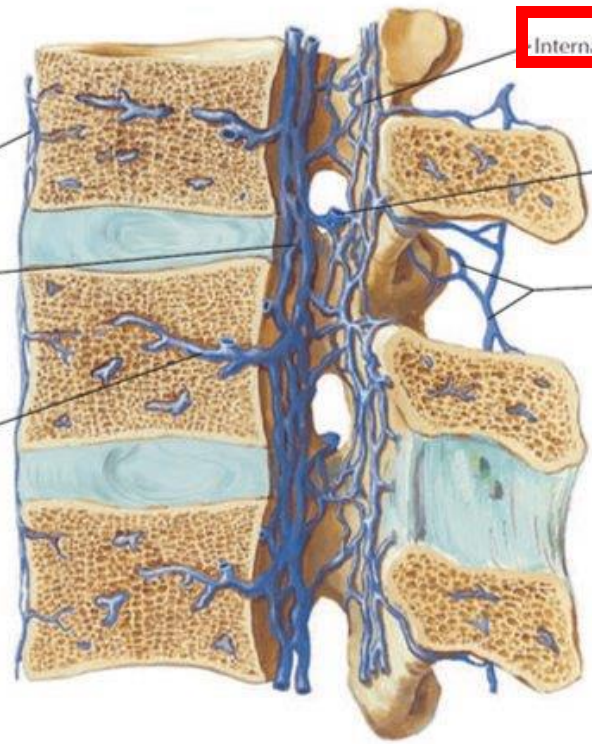
- Anterior and posterior spinal veins drain spinal cord
- Spinal veins drain into internal vertebral venous (Batson's) plexus
- Batson's Plexus is in epidural space
- Batson's plexus communicates with pelvic veins, Dural venous sinuses of cranial vault, and external vertebral venous plexus
- Can serve as collateral pathway for venous flow to systemic veins (caval & azygos veins)





Venous Drainage of Spinal Cord (Batson's)

- Valveless veins located in the epidural space
- Connected to venous vasculature of the pelvis
- Metastasis of pelvic organ cancers to CNS and the vertebral column
- Spread of infection from pelvic region to CNS and vertebral column

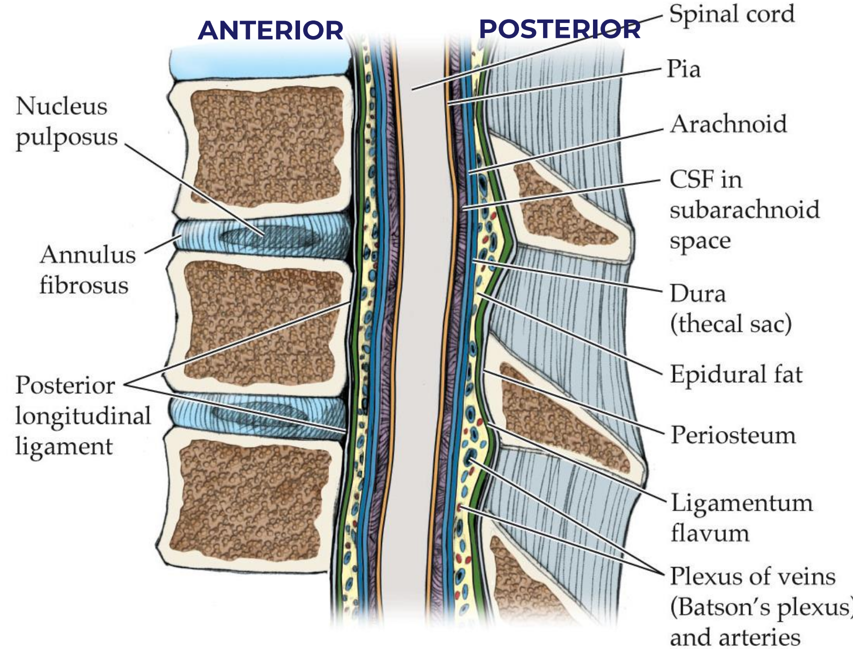


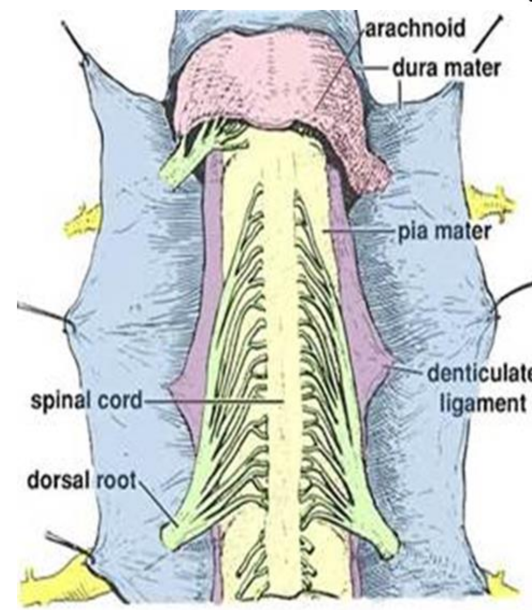
Meninges and Spaces

- Membranes surrounding brain and spinal cord
  - Dura Mater
  - Arachnoid Mater
  - Pia Mater
    - Derived from neural crest cells
    - Creates **Denticulate Ligament**
    - Creates **Filum Terminale**

- Spaces created between membrane
  - Epidural Space → Contains nerves, Batson's plexus
  - Subarachnoid space → Contains Cerebrospinal Fluid (CSF)

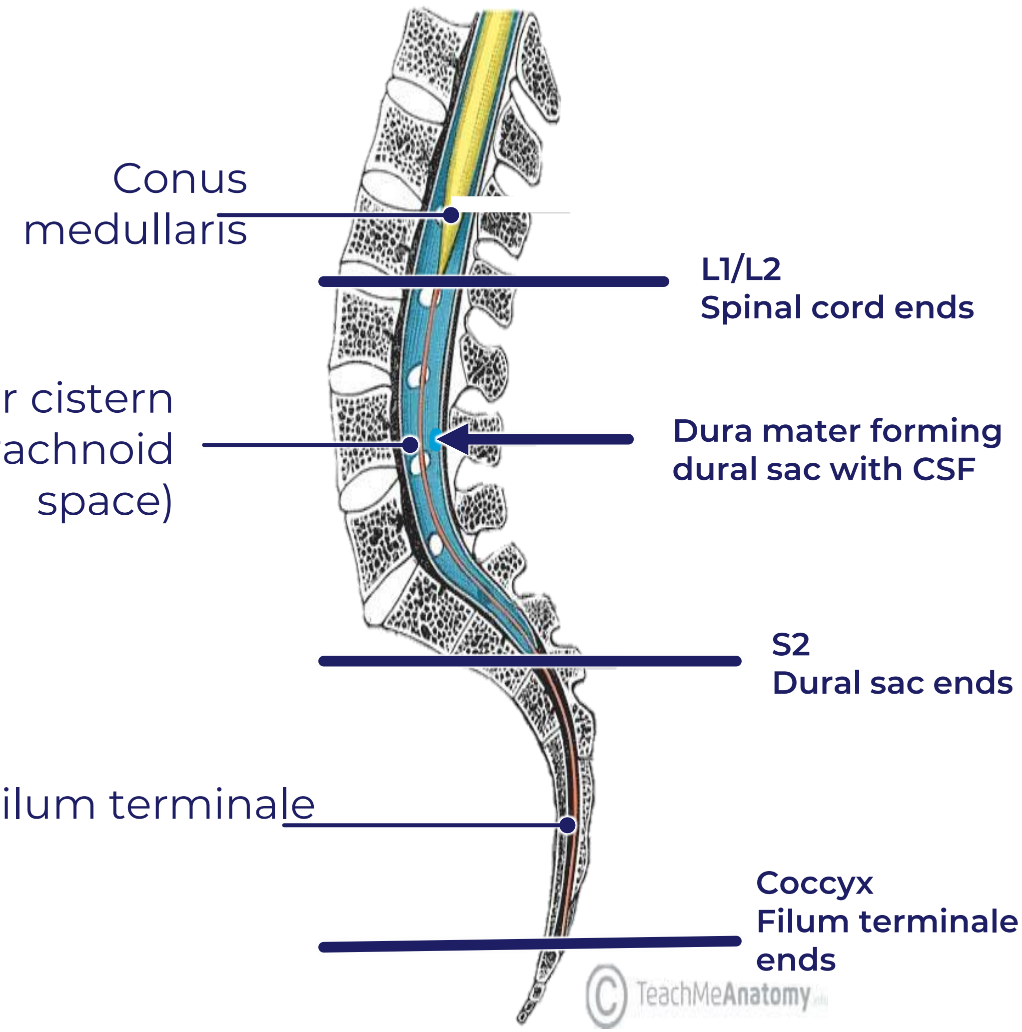
LATERAL VIEW of SPINAL CANAL





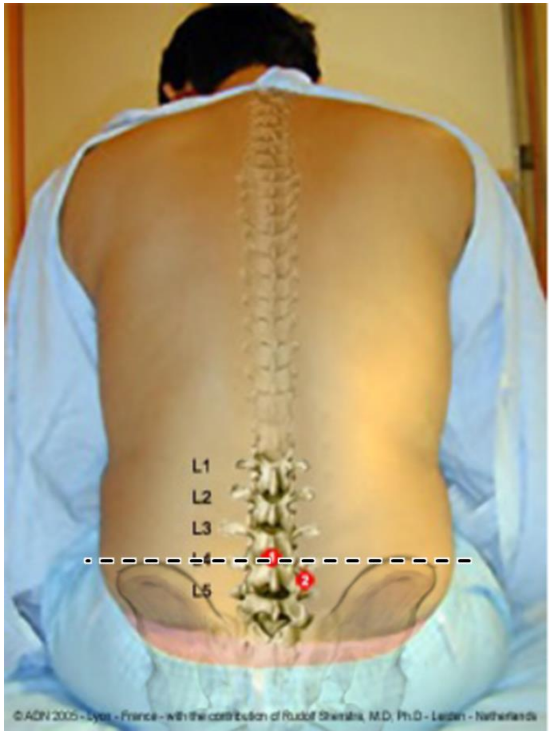
The Dural Sac

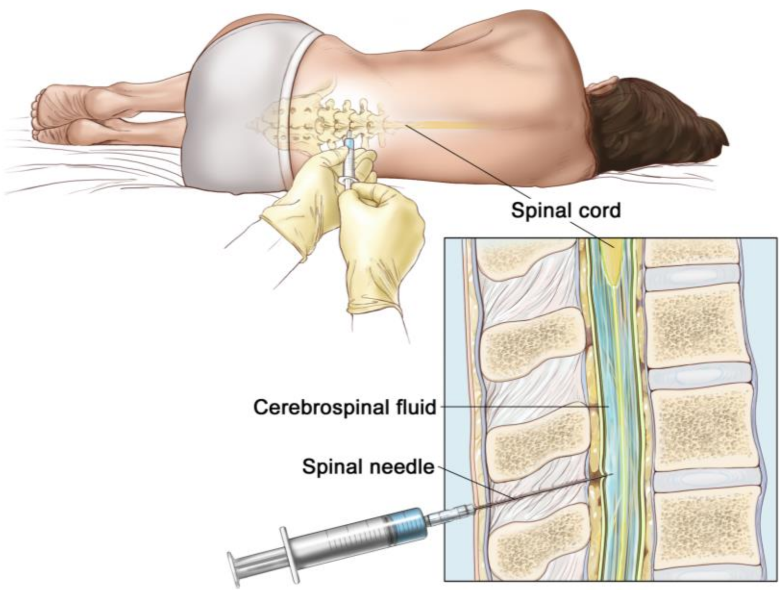
- Spinal cord ends at a higher the vertebral level compared to meninges.
- CSF accumulation caudal to termination of spinal cord → Lumbar cistern
- Lumbar cistern contains filum terminal and cauda equina.



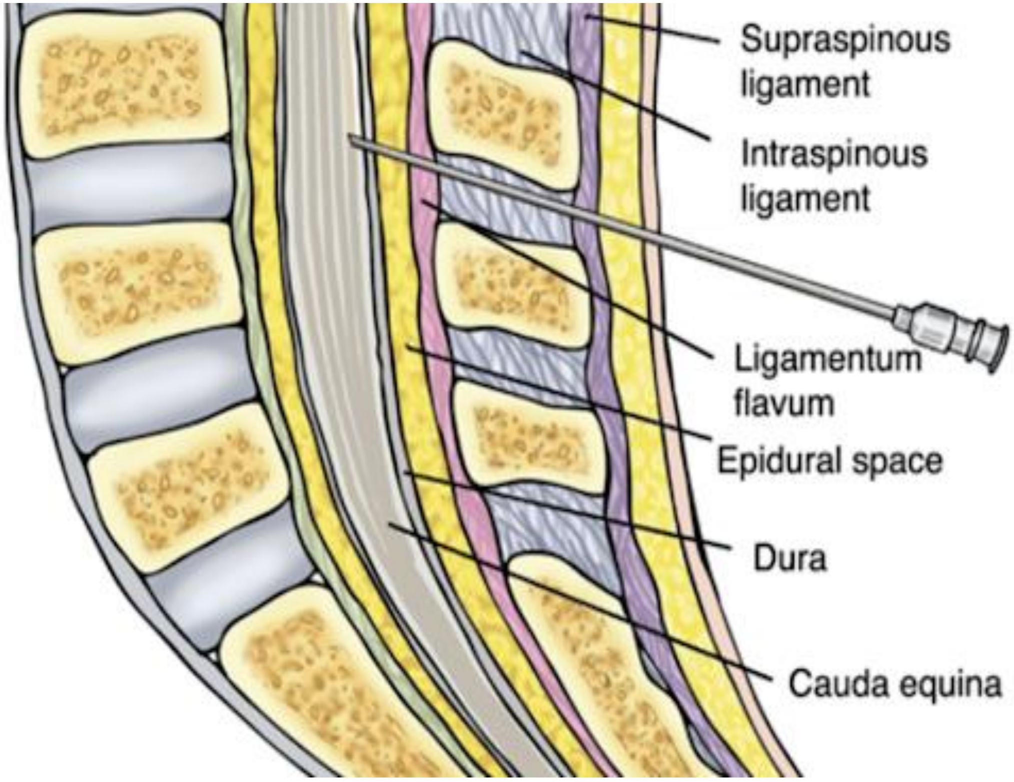
Lumbar Puncture

- Procedure to draw out CSF from lumbar cistern for diagnostic and/or therapeutic reasons
    - Site in an adult – **between L3/L4 or L4/L5**
    - Site in children – **between L4/L5**
- Technique approaches – median or paramedian
- Skin over lower lumbar vertebrae is anesthetized
- Lumbar puncture needle is inserted (cephalad) in midline or paramedian between spinous processes of L4 and L5 at supracristal line.
    - **Supracristal line** – palpate for the top of the iliac crest and imagine a horizontal connecting the two crests.
- After 4-6cm needle “pops” through **ligamentum flavum** and then “pops” again when penetrating the **dura mater** to enter lumbar cistern
- Lumbar puncture is not performed if patient has signs of increased intracranial pressure





Lumbar Puncture Layers



SAGITTAL SECTION

Midline approach

1. Skin
2. Subcutaneous tissue
3. Supraspinous ligament
4. Interspinous ligament
5. Ligamentum flavum
6. Epidural space
7. Dura mater
8. Arachnoid mater
9. Subarachnoid space

Questions email:

ask-anat@sgu.edu